



ScienceDirect®

Optics & Laser Technology

Date: April 2026

[Show more](#) ✓[Research article](#) [Full text access](#) [Get rights and content](#) ↗[Full length article](#)

Comparative investigation of spatter control mechanisms in pure copper welding with 1030nm and 515nm wavelength ring-core adjustable lasers

[Peng Liu^{a, b}](#), [Yan Shi^{a, b}](#) , [Yang Chen^c](#), [Yuan Liu^d](#) [Show more](#) ✓[View PDF](#)[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.optlastec.2026.114718](#) ↗[Article](#)[Recommended articles](#)[Metrics](#)

Abstract

Due to the inherent properties of copper, such as its high reflectivity and thermal conductivity, pure copper laser welding suffers from poor melt pool stability and severe spatter. ring-core adjustable (RCA) laser welding combines the advantages of both Gaussian and ring lasers, demonstrating considerable promise for enhancing molten pool stability and the quality of pure copper welds. This study conducts the first comparative investigation into the mechanisms governing the quality of T2 pure copper welded joints under RCA infrared laser (1030nm) and RCA green laser (515nm) irradiation with various energy distributions. By adjusting the core energy ratio (50%-80%), we systematically

Outline

[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experimental details](#)[3. Results and discussion](#)[4. Analysis of the splashing formation me...](#)[5. Conclusions](#)

analyzed the weld morphology, mechanical properties, and spatter behavior. Through high-speed imaging combined with finite element simulation, the influence of molten metal flow on spatter generation during laser welding was revealed. The results indicate that copper exhibits higher absorption of green light than infrared, forming a wider and deeper melt pool with a larger keyhole opening, which reduces the effect of metal vapor on humping. The V-shaped melt pool with a high width-to-depth ratio helps decrease the vertical flow velocity component. Additionally, compared to infrared, green light absorption is less affected by temperature, resulting in more stable melt pool flow and reduced molten metal impact, thereby suppressing welding spatter. However, green laser welding has slower speed and prolonged melt pool duration, leading to grain coarsening and inferior mechanical properties compared to infrared laser welding.

Keywords

Ring-core adjustable infrared laser; Ring-core adjustable green laser; Copper alloy; Molten pool flow field; Spatter suppression

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

With the rapid development of electric vehicles, power electronics, and new energy industries [1], copper has emerged as a core material for power batteries, energy storage systems, and thermal management components due to its exceptional electrical and thermal conductivity. Consequently, the welding of pure copper is receiving growing attention [2]. Laser welding offers several advantages, including high efficiency, a narrow heat-affected zone, and minimal welding-induced deformation [3], making it particularly suitable for copper sheet welding. However, the inherent material properties of copper alloys, including high thermal conductivity, high reflectivity [4], as well as insufficient

[CRediT authorship contribution statement](#)

[Declaration of competing interest](#)

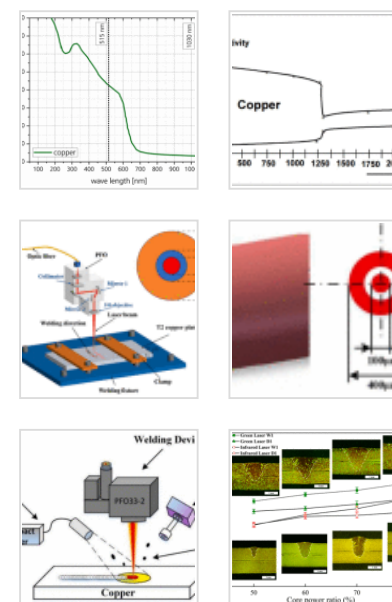
[Acknowledgment](#)

[Data availability](#)

[References](#)

[Show full outline](#) 

Figures (17)



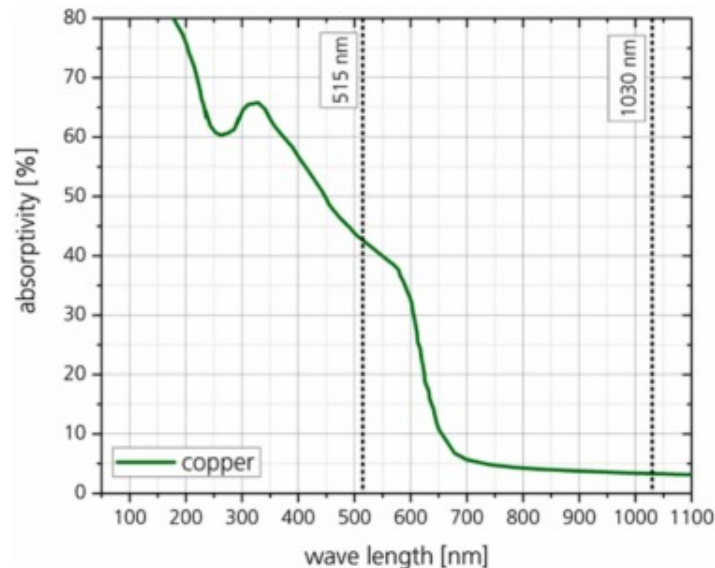
[Show 11 more figures](#) 

Tables (8)

 Table 1. Materials composition of T2 cop...

 Table 2. Parameters of the laser beam so...

initial absorption of infrared lasers and abrupt absorption changes upon melting [5], pose significant challenges to conventional laser welding techniques. Currently, the vast majority of laser welding applications employ infrared laser sources with wavelengths spanning 1030nm to 1070nm [6]. As illustrated in Fig. 1, copper exhibits merely 3%–5% absorptivity to infrared lasers at room temperature, resulting in low energy utilization efficiency, severe laser reflection, a narrow processing window, and susceptibility to defects such as spatter and molten pool instabilities during welding [7].



Download: [Download high-res image \(139KB\)](#)

Download: [Download full-size image](#)

Fig. 1. Absorptivity of copper as a function of wavelength at room temperature [29].

In recent years, green lasers (515~532nm) have emerged as a key technology to address these issues due to their high absorption rate in copper materials [8]. Research has substantiated their distinct advantages in copper alloy welding: A comparative study by Sebastian Engler et al. demonstrated that copper alloys exhibit significantly higher absorptivity for green laser radiation compared to infrared lasers [9]. Furthermore, not only do process parameters significantly influence spatter formation, but variations in

Table 3. Parameters of the programmabl...

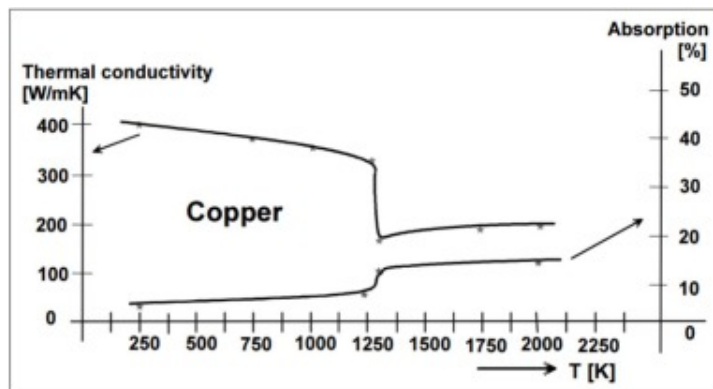
Table 4. Scoring criteria for mass loss an...

Table 5. Orthogonal Test Design for Ring...

Table 6. Orthogonal test design for ring-...

[Show 2 more tables](#) ▼

thermal conductivity and absorptivity during the welding process also play a critical role. Under the premise that other conditions (such as process parameters) are optimally controlled, a smaller solid–liquid absorptivity difference can become a factor conducive to process stability. In Fig. 2, when copper transitions from solid to liquid, its thermal conductivity drops markedly [10]. Moreover, as temperature increases, copper's absorptivity to infrared light rises, while its absorptivity to green light decreases [9], [11], [12]. This leads to easier localized heat accumulation during infrared laser welding, rapidly elevating temperature. Such localized heat buildup further destabilizes the keyhole, thereby increasing spatter. When employing green lasers for copper alloy welding, Martin Haubold et al. observed a pronounced influence of process parameters on weld spatter formation [13]. Research specifically targeting thin copper sheets by Sophie Grabmann et al. confirmed the superior performance of green lasers over infrared lasers for such thin-section welding applications [14]. Additionally, Axel Hess et al. proposed a novel approach by combining high-power infrared light (1030 nm) with low-power green light (515 nm) [15]. In this configuration, the green laser serves to preheat the material, effectively lowering the threshold for deep-penetration welding.



Download: [Download high-res image \(49KB\)](#)

Download: [Download full-size image](#)

Fig. 2. The temperature-dependent behavior of the thermal conductivity of pure copper. [10].

Optimizing the laser beam power density distribution is also a critical strategy for mitigating defects in copper welding. In laser welding, the Gaussian beam, predominantly used due to its prevalence, often induces molten pool instability owing to its highly concentrated energy density at the center [16]. In contrast, the ring beam minimizes spatter and porosity during welding while enhancing the cooling rate at the solidification stage, thereby refining the grain structure and improving the mechanical properties of the weld; however, it significantly reduces the penetration depth. The adjustable ring-core laser, however, combines the advantages of both Gaussian and ring beams. It features a Gaussian-distributed core laser whose high energy density facilitates greater penetration depth in the joint. Simultaneously, its ring laser component, with a large interaction area, enhances molten pool stability [17]. This configuration better balances the thermofluidic forces within the molten pool, demonstrating great potential as a novel laser welding technology. By distributing energy absorption more uniformly within the keyhole, the ring-core laser effectively stabilizes fluid flow in the molten pool [18], [19], [20], [21], [22]. Quanhong Li et al. found that the ring-core beam reduces fluctuations in keyhole depth, enlarges the keyhole opening, and thereby enhances keyhole stability [23]. Furthermore, researchers have identified that the spatial distribution of laser beam energy density critically influences the state of the molten pool during welding [24], [25].

In summary, both ring-core lasers and green lasers demonstrate significant potential for enhancing molten pool stability and improving the welding quality of copper alloys. Existing research has primarily focused on separately investigating the effects of process parameters and the individual mechanisms of ring-core lasers and green lasers in copper alloy welding. However, a systematic exploration of their synergistic effects or a comparative analysis is notably lacking. To address this gap, this study employs a 1030nm adjustable ring-core infrared laser and a 515nm adjustable ring-core green laser to weld T2 pure copper. Utilizing a suite of experimental and simulation-based characterization methods, we systematically and comprehensively investigate the distinct influences of these two laser types on weld morphology, mechanical properties, spatter formation, and molten pool dynamics. The research aims to elucidate the mechanisms by which

adjustable ring-core lasers of different wavelengths control spatter and influence the molten pool flow field, thereby providing critical insights for the application of adjustable ring-core laser welding technology to copper alloys.

2. Experimental details

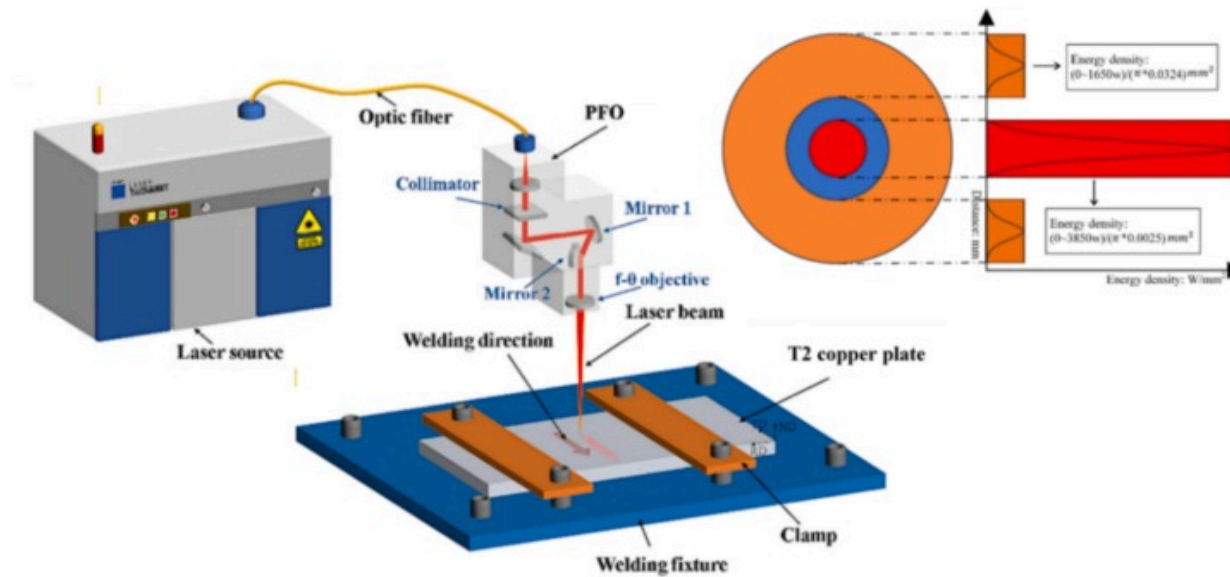
2.1. Experimental materials

Laser lap welding was performed on T2 copper plates measuring 80mm (length)×40mm (width)×1 mm (thickness), with a 20mm overlap region. The chemical composition of the T2 copper alloy is provided in Table 1. Prior to welding, the copper plate surfaces were polished with 800-grit sandpaper to ensure secure clamping during the welding process. The polished plates underwent ultrasonic cleaning in anhydrous ethanol for 15 min, followed by thorough drying using lint-free wipes. Subsequently, the plates were vacuum-sealed in storage bags for subsequent use.

Table 1. Materials composition of T2 copper alloy (wt.%).						
Cu	Bi	Sb	As	Fe	Pb	S
> 99.90	< 0.001	< 0.002	< 0.002	< 0.005	< 0.005	< 0.005

2.2. Experimental equipment

The laser welding system is shown in Fig. 3, which integrates three primary components: a laser pump source, 2-in-1 fiber optics, and programmable focusing optics. The generated beam is directed from the pump source to a beam-splitting module for precise energy ratio distribution. Subsequently transmitted via the 2-in-1 fiber, the beam reaches the programmable focusing optics that govern the final beam delivery path during welding.



Download: [Download high-res image \(190KB\)](#)

Download: [Download full-size image](#)

Fig. 3. Schematic of laser welding setup [30].

The experiment utilized the TruDisk 6001 disk laser and TruDisk 3022 disk laser from Trumpf GmbH (Ditzingen, Germany) as radiation sources, with maximum continuous output powers of 6kW (wavelength λ =1030nm) and 3kW (wavelength λ =515nm), respectively. The performance parameters of the laser welding system and the optical system configuration are detailed in Table 2.

Table 2. Parameters of the laser beam sources and the optical setups used within this work.

Parameter	Unit	Trumpf TruDisk 3022	Trumpf TruDisk 6001
Laser medium	—	Disc	Disc
Wavelength λ	nm	515	1030

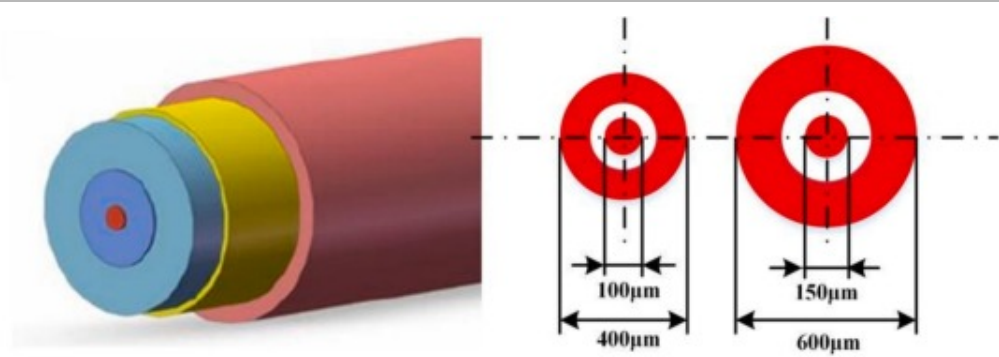
Parameter	Unit	Trumpf TruDisk 3022	Trumpf TruDisk 6001
Laser power $P_{\max}$	W	3000	6000
Fiber diameter d_{LLK}	μm	150	150
Focal diameter d_F	μm	257.1	255.8

The laser beam is emitted from the source, undergoes power conditioning, and is coupled into a 2-in-1 optical fiber. Following this, the beam propagates through the delivery system to a programmable focusing optic (Trumpf PFO 33-2), which performs precise beam focusing along the X- and Y-axes. The key technical parameters of programmable focusing optics and spot diameter for infrared and green lasers are summarized in [Table 3](#). The beam delivery fiber coupled with the Trumpf PFO 33-2 optical system incorporates TRUMPF GmbH's "BrightLine" laser fiber technology, featuring coaxial monolithic fiber architecture for direct-output beam transmission. The "BrightLine" fiber employs a coaxial 2-in-1 architecture that enables dynamic allocation of laser energy between its inner and outer fiber cores. For infrared laser welding trials, a coaxial 2-in-1 fiber with a 100 μm inner core and 400 μm outer ring was implemented. Green laser welding experiments employed a corresponding fiber architecture featuring a 150 μm inner core and 600 μm outer ring. The coaxial fiber configuration and core geometries are illustrated in [Fig. 4](#).

Table 3. Parameters of the programmable focus optics.

Laser type	Fiber diameter (inner/outer) μm	Collimation focal length (μm)	Focusing focal length (μm)	Spot diameter (inner/outer) mm
Infrared laser	100/400	150	345	0.23/0.92

Laser type	Fiber diameter (inner/outer)	Collimation focal length ()	Focusing focal length ()	Spot diameter (inner/outer)
Green laser	150/600	150	255	0.255/1.02



Download: [Download high-res image \(108KB\)](#)
Download: [Download full-size image](#)

Fig. 4. Structure of ring-core composite fiber with inner/outer cores.

3. Results and discussion

Welding experiments were conducted using 1030nm and 515 nm laser beams as radiation sources. Optimal parameters for total laser power, welding speed, and defocusing distance were comprehensively determined via orthogonal experimental design. due to the requirements of practical engineering research, the welding method must be of the incomplete penetration type. It should ensure sufficient tensile-shear strength while avoiding significant thermal deformation or penetration points at the base. To eliminate the influence of false welding on tensile-shear strength, the maximum tensile-shear force of the weld is adopted as the evaluation criterion in subsequent experimental analyses. Additionally, to minimize the impact of welding defects on the electrical conductivity of

the pure copper plate, spattering during the welding process should be reduced as much as possible. For this purpose, two evaluation metrics are established: mass loss and maximum tensile-shear force.

To comprehensively assess both mass loss and mechanical performance, scoring intervals are defined for mass loss and maximum tensile-shear force, respectively, enabling a unified scoring system. The specific scoring criteria are presented in [Table 4](#).

Table 4. Scoring criteria for mass loss and maximum tensile shear force.

Evaluation criteria/score	20	40	60	80	100
Loss of mass/mg	>30	20–30	10–20	5–10	<5
Maximum Tensile Shear/N	<300	300–500	500–750	750–1000	>1000

Following the orthogonal experiments, each resultant weld was evaluated by comparing its mass loss and maximum tensile-shear force with the scoring criteria outlined in [Table 4](#). Using weighting factors of 0.4 for mass loss and 0.6 for maximum tensile-shear force, a comprehensive weld score for each parameter set was calculated by integrating the scores from these two-evaluation metrics. The final results are summarized in [Table 5](#) and [Table 6](#).

Table 5. Orthogonal Test Design for Ring-Core Adjustable Infrared Laser.

number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
1	50	5000	130	2	100	20	52
2	50	5250	100	–2	100	20	52
3	50	5500	120	4	100	20	52
4	50	5750	90	0	20	100	68

number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
5	50	6000	110	-4	100	20	52
6	60	5000	120	-2	100	80	88
7	60	5250	90	4	100	20	52
8	60	5500	110	0	20	100	68
9	60	5750	130	-4	100	20	52
10	60	6000	100	2	20	100	68
11	70	5000	110	4	80	80	80
12	70	5250	130	0	60	100	84
13	70	5500	100	-4	100	40	64
14	70	5750	120	2	20	100	68
15	70	6000	90	-2	20	100	68
16	80	5000	100	0	20	100	68
17	80	5250	120	-4	100	40	64
18	80	5500	90	2	40	100	76
19	80	5750	110	-2	20	100	68
20	80	6000	130	4	60	100	84
21	90	5000	90	-4	40	20	28
22	90	5250	110	2	40	100	76
23	90	5500	130	-2	60	100	84
24	90	5750	100	4	60	100	84

number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
25	90	6000	120	0	20	100	68
L1	55.2	63.2	58.4	68			
L2	65.6	65.6	67.2	70.4			
L3	72.8	68.8	68.8	52			
L4	72	68	71.2	72			
L5	68	68	68	71.2			
σ_L	7.08	2.30	4.84	8.36			

Table 6. Orthogonal test design for ring-core adjustable green laser.

number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
1	50	2600	15	-6	100	20	52
2	50	2700	35	3	100	60	76
3	50	2800	55	-3	100	60	76
4	50	2900	25	6	100	40	64
5	50	3000	45	0	100	80	88
6	60	2600	55	3	80	60	68
7	60	2700	25	-3	80	80	80
8	60	2800	45	6	100	40	64
9	60	2900	15	0	40	100	76
10	60	3000	35	-6	100	40	64

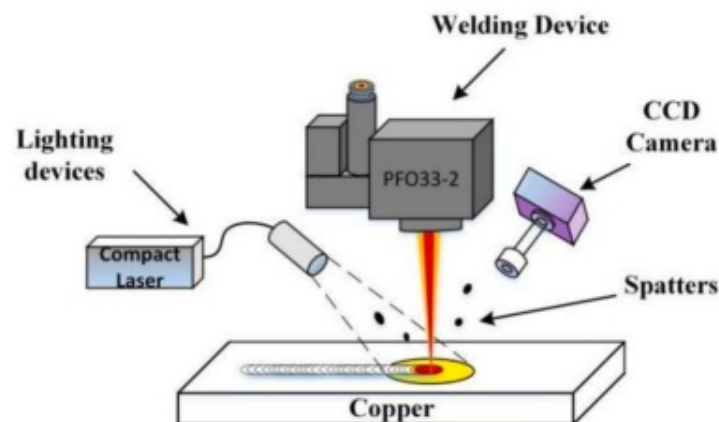
number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
11	70	2600	45	-3	100	100	100
12	70	2700	15	6	60	100	84
13	70	2800	35	0	40	100	76
14	70	2900	55	-6	100	80	88
15	70	3000	25	3	80	100	92
16	80	2600	35	6	80	60	68
17	80	2700	55	0	40	100	76
18	80	2800	25	-6	100	60	76
19	80	2900	45	3	80	100	92
20	80	3000	15	-3	20	100	68
21	90	2600	25	0	20	100	68
22	90	2700	45	-6	100	20	52
23	90	2800	15	3	40	100	76
24	90	2900	35	-3	60	100	84
25	90	3000	55	6	100	100	100
L1	71.2	71.2	71.2	66.4			
L2	82	73.6	81.6	80.8			
L3	80.4	73.6	73.6	76.8			
L4	76	80.8	79.2	81.6			
L5	76	82.4	76	76			

number	Core/%	P/W	V/mm·s ⁻¹	F/mm	Loss of mass	Maximum Tensile Shear Force	Point
σ_L	7.03	4.95	4.18	6.06			

Under infrared laser conditions, the core energy ratio L3 yields the highest value, indicating that the third level corresponds to a higher average quality of welded joints. Therefore, a core energy ratio of 70% can be considered the optimal choice. Similarly, the laser total power of 5500W, the welding speed V of 120mm/s, and the defocus amount F of +2mm are identified as preferable settings.

Under green laser conditions, the optimal core energy ratio is determined to be 60%, with the laser total power of 3000W, welding speed V of 25 mm/s, and defocus amount F of +3mm being the corresponding optimal parameters. Subsequently, single-factor experiments were conducted targeting the core energy ratio based on the optimal parameters.

Furthermore, the welding process was observed via high-speed imaging to investigate the effects of adjustable ring-core laser on weld performance and defects. The high-speed camera was positioned according to the configuration shown in Fig. 5. The camera operated at a frame rate of 10,000 frames per second (fps) with a resolution of 1280×600. It was used to capture the welding process, particularly recording a 100-millisecond time interval to analyze the fluctuation behavior of the molten pool. Based on these observations, the frequency of keyhole collapse per unit time was calculated.



[Download: Download high-res image \(49KB\)](#)

[Download: Download full-size image](#)

Fig. 5. Schematic diagram of high-speed camera positioning.

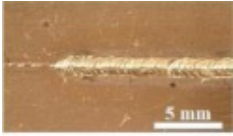
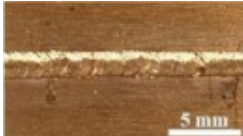
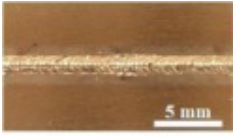
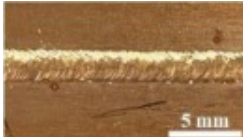
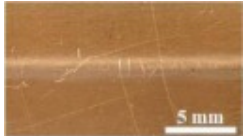
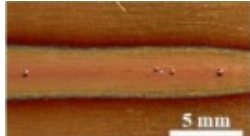
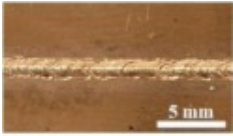
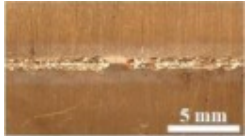
The weld formation quality analysis of pure copper plates in this study was conducted primarily through macroscopic surface morphology, cross-sectional morphology, mass loss quantification, and tensile-shear testing. The mass difference before and after welding was utilized as a quantitative indicator to evaluate spatter generation during the welding process. By establishing a mass defect-based assessment system for laser welding imperfections, this approach achieved quantitative analysis of welding defects.

3.1. Weld morphology

First, an analysis was conducted on adjustable ring-core infrared laser welding. Based on the results of orthogonal experiments, the total laser power was set at 5500W, the welding speed at 120mm/s, and the defocusing amount at +2mm as constants, while the core energy ratio was varied at 50%, 60%, 70%, and 80%. As shown in Table 7, when the core energy ratio was 50%, the power density at the beam center was relatively low. Combined with the high reflectivity of copper to infrared lasers, most of the laser energy was reflected during the welding process, resulting in poor weld seam continuity and

ineffective welding, with almost no observable changes on the backside of the base material. As the core energy ratio increased to 60%, the metal melting area expanded, but the fish-scale patterns became coarse and unevenly spaced. Simultaneously, a slight white heat-affected zone appeared on the backside of the base material. When the core energy ratio reached 70%, the weld seam exhibited continuous and uniform morphology on the front side, while the visibility and consistency of the white traces on the backside of the weld became more pronounced. When the core energy ratio was further increased to 80%, significant fluctuations in the solidification morphology ahead of the weld pool were observed. At this stage, the backside of the weld was nearly completely melted.

Table 7. Weld morphology under different core energy ratios.

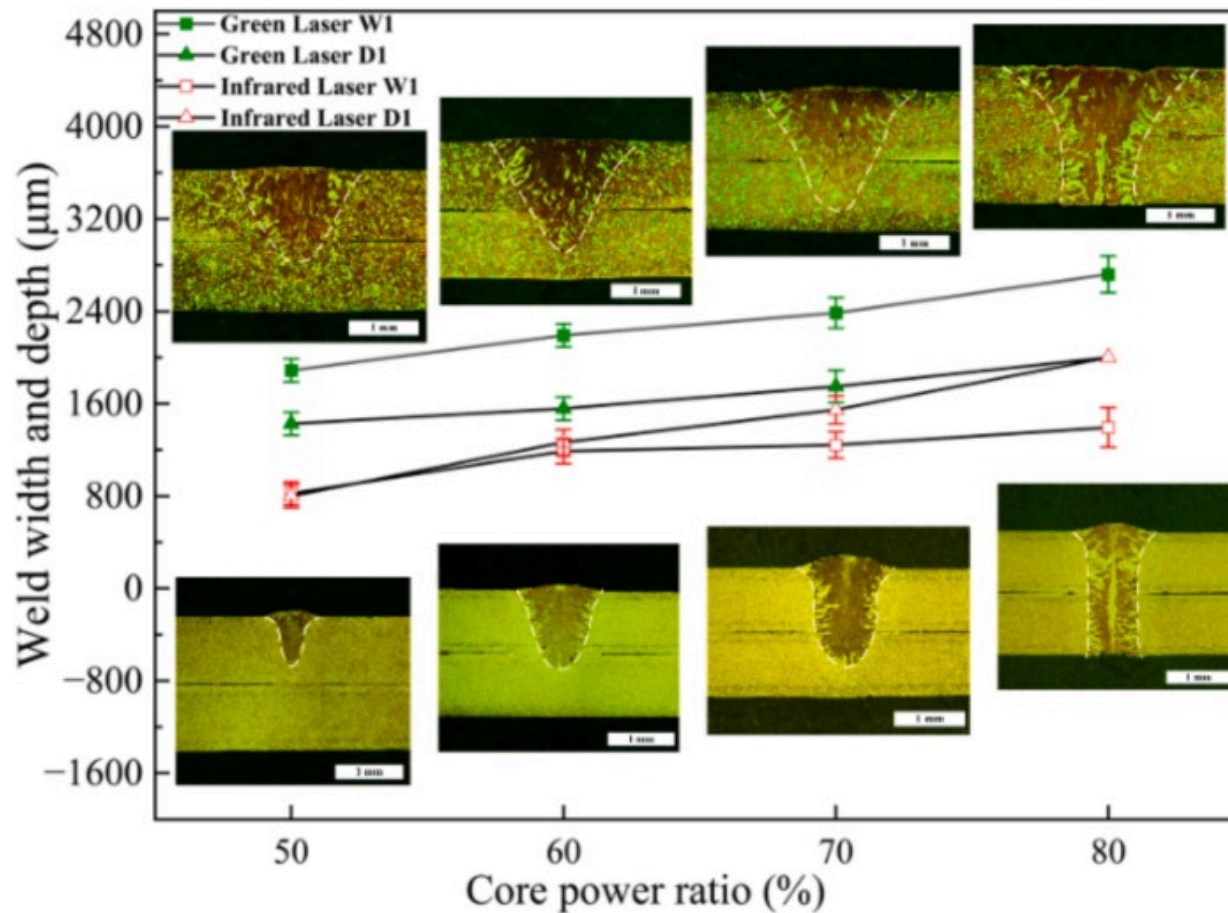
Core/%	Surface-IR	Bottom-IR	Surface-G	Bottom-G
50				
60				
70				
80				

For green laser welding with adjustable ring-core power distribution, based on the orthogonal experimental results, the total laser power was set at 3000W, the welding speed at 25mm/s, and the defocusing distance at a constant+3mm. The core energy power ratios were set at 50%, 60%, 70%, and 80%. At a 50% core energy ratio, only 50% of the total laser power was allocated to the core beam, resulting in lower power density and weaker interaction between the laser and the laser-induced plasma. The weld surface exhibited smooth and regular formation with almost no spatter. However, due to insufficient energy input, the backside of the weld showed poor uniformity. When the core energy ratio increased to 60%, the power density became more moderate. The weld surface widened, and the “fish-scale” pattern appeared clear and uniform without noticeable fluctuations. Due to the increased energy input, the backside oxidation zone expanded but remained uniform, resulting in overall better welding quality. At a 70% core energy ratio, the power density was relatively high, leading to increased spatter on the weld surface and excessive molten pool fluidity. This indicated an imbalance in surface tension, causing uneven weld ripples. The backside of the weld exhibited partial penetration points, with some areas showing full penetration, suggesting that the laser energy input reached the critical threshold for keyhole welding. The welding mode gradually transitioned from non-penetration to partial penetration. At an 80% core energy ratio, the power density far exceeded the optimal level, resulting in severe spatter on the weld surface, rough and irregular fish-scale patterns, and complete penetration on the backside. During green light welding, the relatively slow welding speed prolongs the existence of the molten pool, consequently extending the duration of the high-temperature liquid metal and increasing its exposure to the ambient atmosphere, thereby exacerbating oxidation.

It can be observed that the infrared laser weld exhibits its optimal surface morphology at a 70% core energy ratio, while the green laser weld achieves its best appearance at a 60% ratio.

3.2. Morphology of molten pool cross-section

The cross-sectional shape of the weld seam is closely related to the laser beam's energy input and absorption rate. During the welding process, variations in the core energy ratio alter the energy distribution of the laser beam, thereby affecting its energy density and total heat input. These changes ultimately lead to modifications in the weld seam's cross-sectional morphology, as illustrated in Fig. 6.



[Download: Download high-res image \(533KB\)](#)

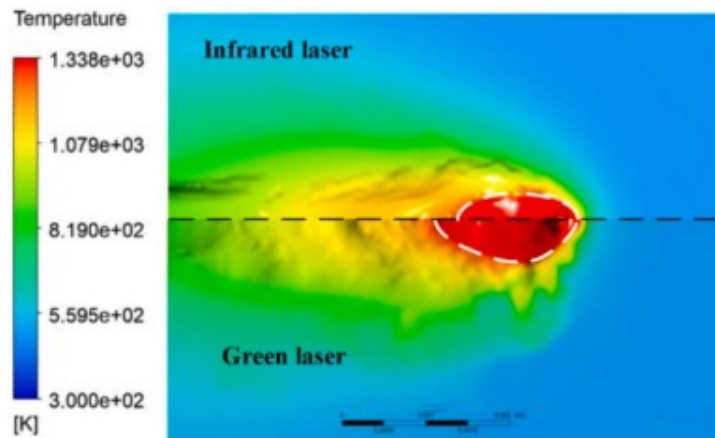
[Download: Download full-size image](#)

Fig. 6. Weld cross-section dimensions under different core energy ratios.

We define the penetration depth as $D1$ and the weld width as $W1$. With other parameters held constant, both the penetration depth ($D1$) and the weld width ($W1$) exhibit a corresponding increase as the core energy proportion rises.

Upon examining the cross-sectional morphology of the molten pool, it was observed that under non-penetration welding conditions, the width-to-depth ratio of the molten pool exceeds 1 when using green laser welding, whereas the ratio falls below 1 in the case of infrared laser welding.

In the horizontal direction, as shown in Fig. 7, the wider molten pool observed in green laser welding can be attributed to the larger spot size, slower processing speed, and higher absorptivity.



[Download: Download high-res image \(56KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Comparison of Spot Sizes between Infrared and Green Lasers.

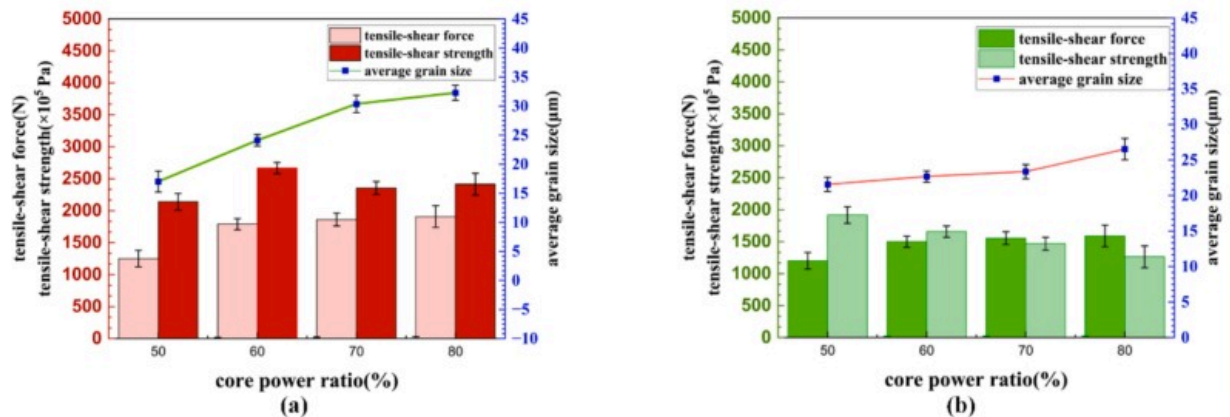
In the vertical direction, the absorptivity of copper alloy for green laser is approximately 40% at room temperature, decreasing to about 25% with rising temperature [12]. In contrast, its absorptivity for infrared laser increases from 5% to 10% as the temperature rises[9], [11]. This results in welds produced with the green laser being more prone to

transverse growth (widening), while those produced with the infrared laser are more prone to longitudinal growth (deepening).

In summary, the combined differences in the horizontal and vertical directions are the primary reasons why the cross-sectional width-to-depth ratio of green laser weld joints exceeds 1, whereas that of infrared laser weld joints is less than 1.

3.3. Mechanical properties

To investigate the effect of core-to-ring ratio on the mechanical properties of welds, tensile-shear tests were performed on welded joints with different core-to-ring ratios while maintaining both laser wavelengths at their respective optimal parameters. The results are shown in Fig. 8(a) and 8(b). Furthermore, grain size was measured in accordance with the ASTM E112 international standard using a Leica DM2700 M optical microscope.



[Download: Download high-res image \(246KB\)](#)

[Download: Download full-size image](#)

Fig. 8. Mechanical properties and average grain size: (a) Infrared laser; (b) Green laser.

For the ring-core adjustable infrared laser welding, as the core energy ratio increased, the effective bonding area gradually expanded, and the maximum tensile-shear force showed

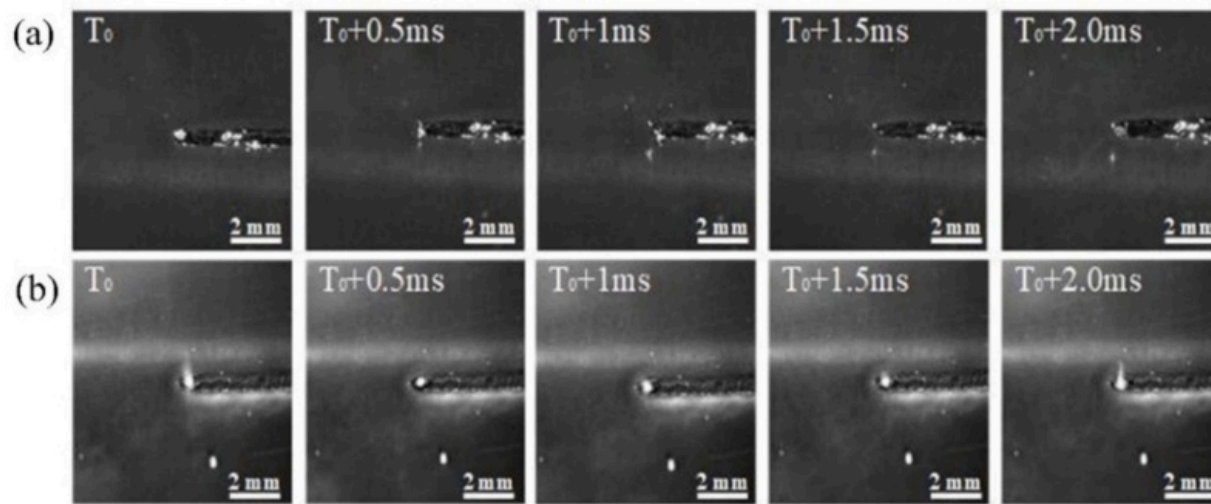
a continuous upward trend, though the rate of increase slowed down after the core power proportion reached 70%. The maximum tensile-shear strength first increased and then decreased with increasing core power proportion, reaching a peak value of 266.722 MPa at 70% core power. Further increasing the core-to-ring ratio resulted in decreased tensile strength. This is due to the higher power density of the beam, which results in an elevated molten pool temperature, extended solidification time, and favorable conditions for grain growth, ultimately leading to grain coarsening.

For the ring-core adjustable green laser welding, as the core energy ratio increased, the maximum tensile-shear force of the welded joints exhibited an increasing trend. At 50% core power, due to the relatively low laser power allocated to the core beam and the resulting low power density, the molten metal zone within the weld pool was limited during laser-material interaction. This made it difficult to form sufficient effective bonding areas and often led to false welding defects, resulting in the lowest maximum tensile-shear force. At 60% and 70% core power, the increased laser power density allowed nearly complete energy absorption through multiple reflections of the laser beam inside the keyhole. As the core-to-ring ratio increased, the molten metal zone within the weld pool expanded, leading to larger effective bonding areas and consequently an upward trend in maximum tensile-shear force. However, with further increases in the core-to-ring ratio, the maximum tensile-shear strength of the welded joints showed a decreasing trend. This was because more efficient energy absorption within the molten pool prolonged its existence time, providing favorable conditions for grain growth. The resulting coarsening of the grain structure in the welded joints led to a reduction in tensile-shear strength.

3.4. Molten pool spatter

To clarify the influence of infrared laser and green laser on molten pool spatter behavior, high-speed imaging was employed to observe the molten pool state during welding of T2 pure copper plates with different laser wavelengths. For both wavelengths, optimal process parameters yielding better welding quality were selected as stable welding cases

for comparative analysis between infrared and green laser welding of pure copper plates, as shown in Fig. 9, During ring-core adjustable infrared laser welding, due to the high-speed welding conditions, the flow behavior of molten metal inside the pool was more intense, and keyhole closure occurred at a higher frequency. In contrast, when using ring-core adjustable green laser welding, the keyhole remained in a consistently open state, and the molten pool exhibited a more stable flow condition.



[Download: Download high-res image \(131KB\)](#)

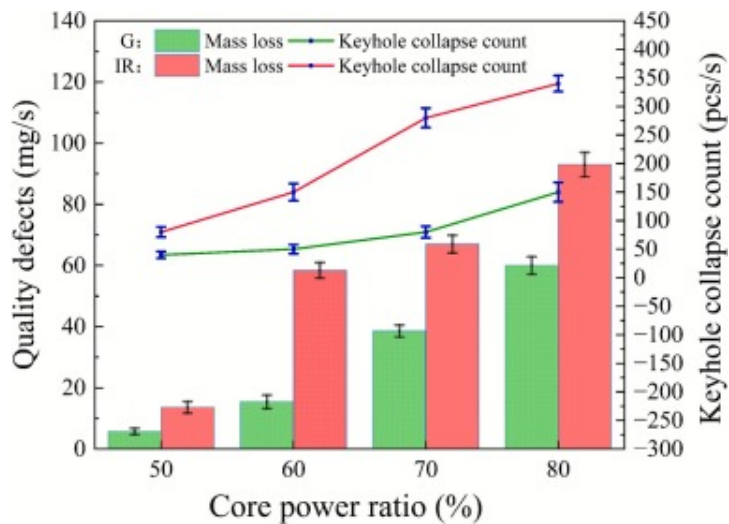
[Download: Download full-size image](#)

Fig. 9. Molten pool and spatter behavior laser welding: (a) Infrared laser; (b) Green laser.

Due to equipment limitations, the maximum output power of the green laser is restricted to 3kW, which prevents it from achieving high-speed welding. The experimental welding speed for the green laser is 25m/s, whereas the infrared laser achieves a welding speed of 120m/s. It's 4.8 times faster than the green laser. Consequently, the duration of welding with the green laser is also 4.8 times longer than that with the infrared laser. Therefore, the mass loss per unit time is used to evaluate the spatter generation for both laser wavelengths. It is acknowledged that the measured mass difference may include a minor contribution from elemental vaporization; however, given the magnitude of the loss and

its strong correlation with observed spatter behavior, it is considered a robust comparative metric for spatter generation in this study [26], [27].

By measuring the mass loss and observing the welding process via high-speed imaging, the mass loss per unit time and the keyhole oscillation frequency during the welding of T2 pure copper plates with both laser wavelengths were statistically analyzed, as shown in Fig. 10.



Download: [Download high-res image \(196KB\)](#)

Download: [Download full-size image](#)

Fig. 10. Mass loss and keyhole oscillation occurrences during infrared vs. green laser welding.

Under different core energy ratios, the mass loss during infrared laser welding is approximately 3–4 times that of green laser welding. High-speed imaging observations reveal that the keyhole oscillation frequency during infrared laser welding is also higher than that of green laser welding, by about 2–3 times. This indicates that the higher mass loss per unit time during infrared laser welding is attributed to its high-speed welding regime, where the flow behavior of molten metal inside the weld pool intensifies, and the

driving forces within the pool increase, leading to reduced keyhole stability. In this case, spatter formation primarily occurs during keyhole opening and closing stages. As the frequency of keyhole oscillations increases, the collapse of the keyhole becomes more frequent. After collapse, the internal cavity pressure surges, causing molten metal to eject from the opening and form spatter.

In contrast, during ring-core adjustable green laser welding, the higher absorption rate and more stable energy coupling in copper alloys result in smaller weld pool fluctuations and fewer keyhole oscillations compared to infrared laser welding. Consequently, spatter formation is more suppressed, and the mass loss is significantly reduced.

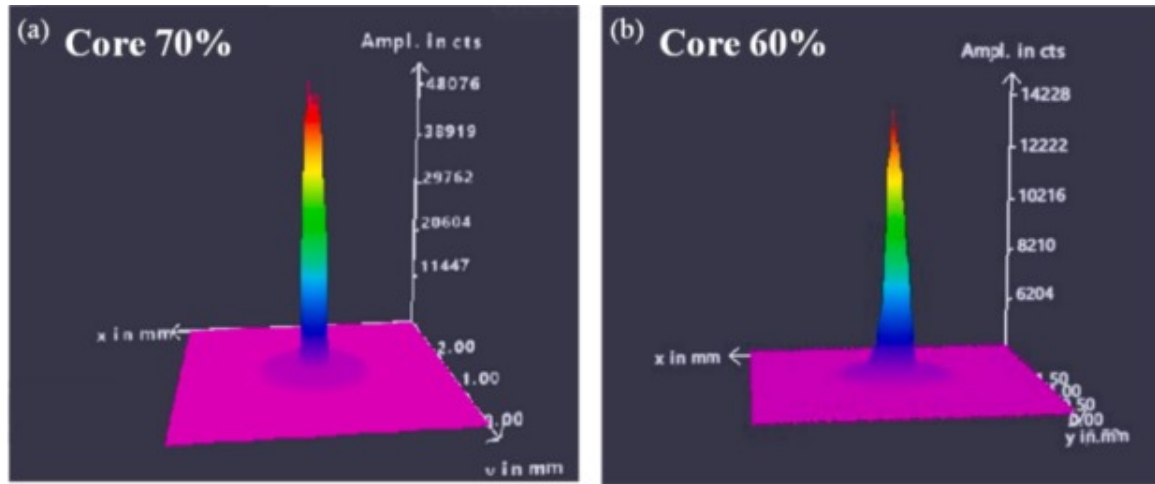
4. Analysis of the splashing formation mechanism

The above welding process images captured by the high-speed camera did not fully reflect the keyhole behavior and molten pool flow in the narrow gap. A hydrodynamic model was developed to simulate the evolution of the molten pool flow and the keyhole dynamic behavior.

4.1. Modeling

a) Heat source model.

The thermal source model significantly influences the accuracy of numerical simulation results for the laser welding process. The ring-core laser heat source consists of two parts: a central point source and an outer ring-shaped source. The energy distribution of the central point source follows a Gaussian distribution, while the energy allocation of the outer ring-shaped source can be considered uniform. To precisely simulate this process, a beam diagnostic device was employed to evaluate the energy distribution of infrared and green laser beams under optimal parameter conditions, as shown in [Fig. 11](#).



Download: [Download high-res image \(51KB\)](#)

Download: [Download full-size image](#)

Fig. 11. Measured intensity distribution of laser beams: (a) Infrared laser; (b) Green laser.

Combined with actual measurement results, the energy density expression of the RCA beam is established as:

$$q_L(r, z) = \begin{cases} \frac{Q_C}{\pi r_c^2} \exp\left(-\frac{r^2}{r_c^2}\right) & (0 < r < r_c) \\ \frac{Q_R}{\pi(r_{R2}^2 - r_{R1}^2)} & (r_{R1} < r < r_{R2}) \end{cases} \quad (1)$$

The energy density of laser irradiated on materials surface $q_L(r, z)$ can be expressed by Eq. (1), where Q_C, Q_R describe the laser power for the core and ring beam. The model parameters r_{R1} and r_{R2} represent the inner and outer radius of the ring at the focusing plane of the ring-light. r_c denotes the spot radius of the core beam at the focal plane.

During the laser deep penetration welding process, there exists a highly complex energy interaction between the laser and metallic materials within the keyhole. The laser beam undergoes multiple reflections inside the keyhole. Energy absorption and reflection occur during each reflection process. The multiple laser reflections inside the keyhole facilitate

the absorption of laser energy by the keyhole, thereby enhancing laser energy utilization efficiency. This study employs the Fresnel energy absorption model in numerical simulations of point-circular laser welding to describe the multiple reflection phenomena of lasers within the keyhole. The Fresnel energy absorption rate is calculated as follows:

$$\alpha(\varphi_n) = 1 - \frac{1}{2} \left(\frac{1 + (1 - \varepsilon \cos \varphi_n)^2}{1 + (1 + \varepsilon \cos \varphi_n)^2} + \frac{\varepsilon^2 - 2\varepsilon \cos \varphi_n + 2(\cos \varphi_n)^2}{\varepsilon^2 + 2\varepsilon \cos \varphi_n + 2(\cos \varphi_n)^2} \right) \quad (2)$$

The angle between the incident direction of the laser beam and the material position in the normal direction denoted as φ_n ; the Fresnel energy absorption rate represented by $\alpha(\varphi_n)$; and the laser-material correlation coefficient is denoted as ε .

b) Control equations.

Laser deep welding involves mass and heat transfer, which is based on the conservation equations of fluid dynamics: the mass conservation equation (continuity equation), momentum conservation equation (Navier-Stokes equation), and energy conservation equation.

(1) Conservation of mass equation

$$\frac{\partial}{\partial t}(\rho) + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad (3)$$

In the given expression, the variables u , v and w represent the fluid velocity components in the x , y and z directions, respectively. The ρ denotes the fluid density, while μ signifies the fluid viscosity coefficient.

(2) Conservation of momentum equation (Navier-Stokes equation)

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = \nabla P + \mu \nabla^2 \vec{v} - \rho K \vec{v} + \vec{f}_b \quad (4)$$

In this context, the variables denoted as P represent pressure, t represent time, $\vec{V}$ represent the velocity vector, K represent the Carman-Kozeny coefficient, and $\vec{f}_b$ represent buoyancy.

(3) Conservation of energy equation

$$\rho \left(\frac{\partial h}{\partial t} + \vec{V} \cdot \nabla h \right) = \nabla \cdot (\kappa \nabla T) \quad (5)$$

Where h represents enthalpy, T denotes fluid temperature, and κ signifies fluid thermal conductivity.

The free surface tracking method known as Volume of Fluid (VOF) was employed to determine the free liquid level within the melt pool. This was achieved by introducing the fluid volume fraction denoted as $F(x, y, z, t)$.

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{V} \cdot F) = 0 \quad (6)$$

During the laser welding process, the material undergoes a transformation from a solid metal to liquid and gaseous states. To accurately depict this process in the numerical simulation summary, the following phase transition model is utilized:

$$h = \begin{cases} \rho_s C_s T (T \leq T_s) \\ h(T_s) + h_{sl} \frac{T - T_s}{T_l - T_s} (T_s \leq T \leq T_l) \\ h(T_l) + h_{sl} \frac{T - T_l}{T_l - T_l} (T_l \leq T) \end{cases} \quad (7)$$

The variables in question are as follows: ρ_s represents the density of the solid phase of the material, ρ_l denotes the density of the liquid phase of the material, C_s signifies the specific heat per unit volume of the solid phase of the material, C_l indicates the specific heat per unit volume of the liquid phase of the material, T_s represents the temperature at the solid-phase line, T_l denotes the temperature at the liquid-phase line, and h_s signifies the latent heat of melting for the solid phase change.

c) Melting Pool Driving Force Model.

The flow behavior inside the molten pool is primarily in dynamic equilibrium under the influence of internal driving forces such as liquid metal evaporation reaction force,

surface tension, and thermal buoyancy. Therefore, it is essential to include the molten pool driving force as a source term in the calculation.

The force of the metal evaporation reaction was numerically calculated using the following empirical equation:

$$P_r = 0.54P_0 \exp\left(L_v \frac{T-T_b}{RT_b}\right) \quad (8)$$

In the given equation, P_0 denotes the atmospheric pressure, L_v signifies the latent heat of vaporization, T_b represents the boiling point of the metal material, and R stands for the gas constant. The equation used to determine the surface tension of the molten pool is expressed as follows.

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_0) \quad (9)$$

Where γ represents the surface tension of the liquid metal at temperature T , γ_0 denotes the surface tension of the liquid metal at its melting point, $\frac{d\gamma}{dT}$ signifies the surface tension gradient, and T_0 corresponds to the boiling point of the material. In this study, the thermal buoyancy within the molten pool is addressed using the Boussinesq approximation in the computational model. The thermal buoyancy equation is presented as follows:

$$F_b = \rho g \beta (T - T_0) \quad (10)$$

Where T_0 represents the ambient temperature; T is the melt temperature; β is the melt thermal expansion coefficient.

d) Material properties.

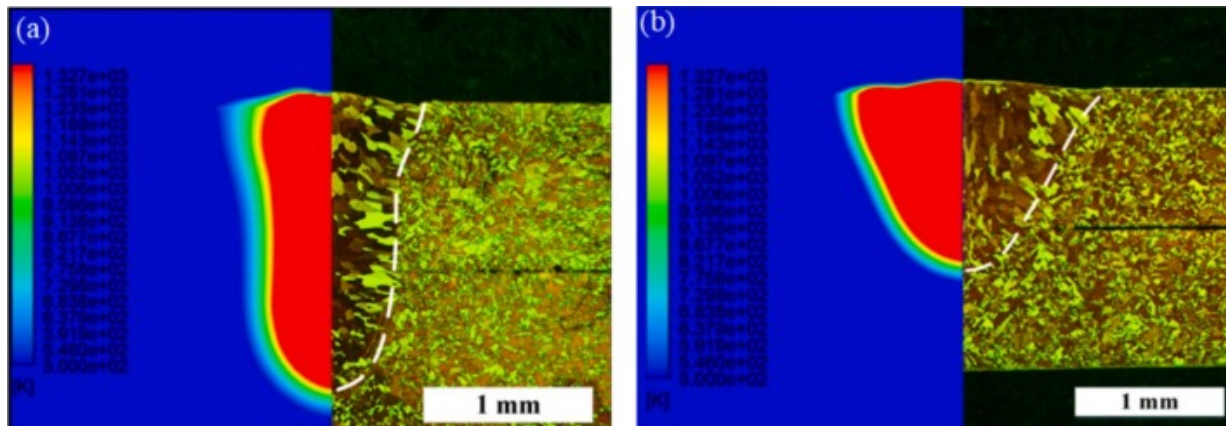
The laser welding experiment utilized Industrial T2 pure copper. The thermophysical characteristics of the material can be found in [Table 8](#).

Table 8. Physical parameters of T2 pure copper.

Parameters	Value	Unit
Density / ρ	8960	$kg \cdot m^{-3}$
thermal conductivity / κ	-0.0603T+415.4 (solid)	$w \cdot m^{-1} \cdot K^{-1}$
	172(liquid)	
specific heat / C_P	0.1482T+327.7 (solid)	$J \cdot kg^{-1} \cdot K^{-1}$
	572.12 (liquid)	
Molten temperature / T_m	1338.75 (solid)	K
	1356.15 (liquid)	
Latent heat of fusion	188,000	$J \cdot kg$
Viscosity / μ	1(solid)	$kg \cdot m^{-1} \cdot s^{-1}$
	$-3.37 \times 10^{-6}T + 8.35 \times 10^{-3}$ (liquid)	
Temperature coefficient of surface tension / γ	-1.5625×10^{-4}	$N \cdot m^{-1} \cdot K^{-1}$

e) Verification of the Heat Source Model.

After establishing the heat source model, the numerically simulated heat-affected zone (HAZ) was compared with the experimental HAZ of the weld cross-section, as shown in Fig. 12(a) and 12(b). The results demonstrate that the simulated HAZs of both the adjustable ring-core infrared laser and the adjustable ring-core green laser exhibit good agreement with the experimental HAZs, confirming the accuracy of the developed heat source model.



Download: [Download high-res image \(190KB\)](#)

Download: [Download full-size image](#)

Fig. 12. Verification of the heat source model: (a) infrared laser; (b) green laser.

4.2. Analysis of spatter formation conditions

(a) Influencing factors of spatter formation

Spattering occurs when a localized portion of the molten material acquires sufficient momentum normal to the melt surface, causing it to detach from the bulk melt. Splatter droplets detach from the weld pool when the vertical component of the near-surface flow exceeds a critical ejection threshold. The escape threshold is established by a combination of the local values of surface tension, surface geometry, viscosity, and gravity. The surface tension is the most important consideration and this can be greatly affected by local temperature and surface chemistry (oxidation or impurities).

The droplet escape condition can be described mathematically by [28]:

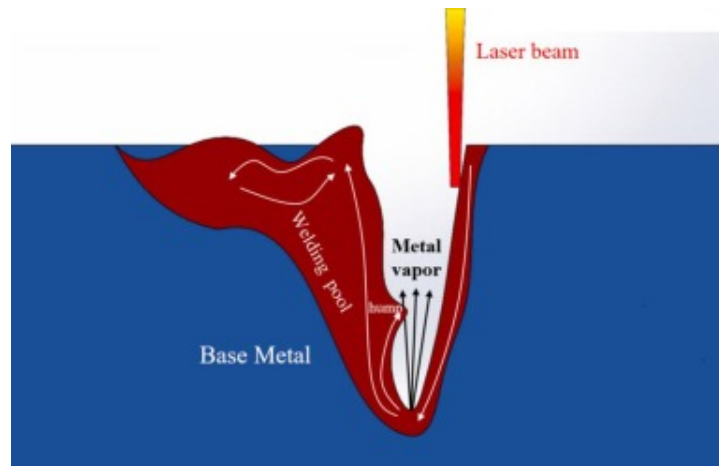
$$\rho v_z^2 > \frac{2\sigma}{R} \quad (11)$$

Where the average melt density is ρ , the vertical velocity component is v_z , and the surface tension is σ . The local curvature radius of the melt is R .

When material properties are determined, the vertical velocity of liquid metal v_z becomes the decisive factor governing whether the melt exceeds the threshold for spattering generation. Accordingly, subsequent analysis will concentrate on the molten pool flow field.

(b) Two distinct spatter formation mechanisms.

During laser welding, the mechanisms for acquiring vertical velocity differ between the molten pool and keyhole regions. As illustrated in Fig. 13:



Download: [Download high-res image \(36KB\)](#)

Download: [Download full-size image](#)

Fig. 13. Two distinct spatter formation mechanisms during ring-core laser welding.

In the molten pool region: Laser irradiation ablates the keyhole front wall, generating metallic vapor. The recoil pressure from this vapor imparts initial kinetic energy to the molten metal, driving it to flow around the keyhole toward the rear pool. This upward flow impacts the pool surface behind the keyhole wall.

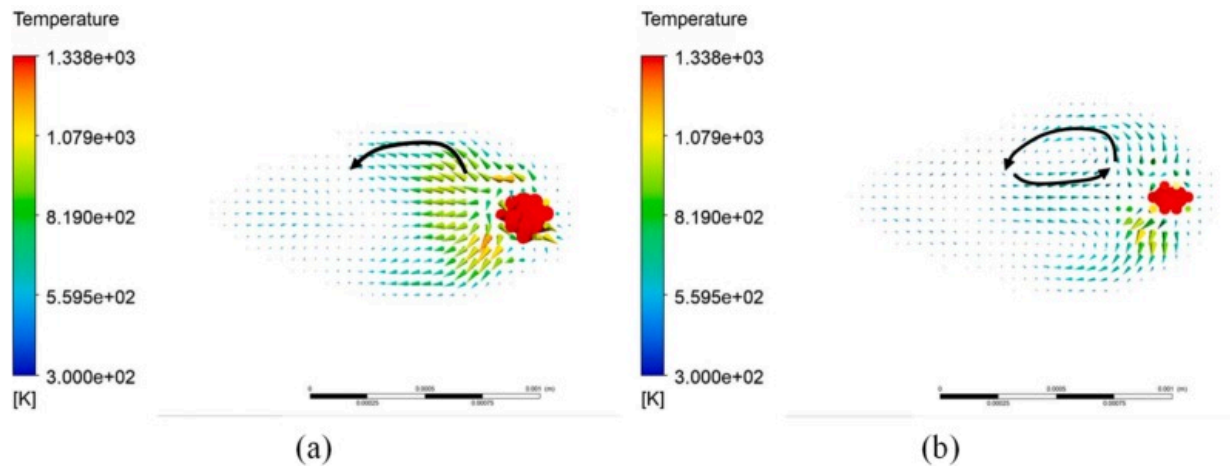
In the keyhole region: Fluctuations in the keyhole cause upward-moving vapor (produced by metal evaporation at the keyhole base) to exert a vertical force on the hump formed by

molten metal impacting the rear wall. This force generates an upward velocity component, ejecting the hump from the pool as spatter.

4.3. Molten pool region spatter formation and suppression mechanism

(a) Top surface flow field of molten pool.

The flow patterns of molten metal on the top surface of the molten pool during welding of pure copper plates using ring-core adjustable infrared laser and ring-core tunable green laser are shown in Fig. 14(a) and 14(b), respectively. Compared with the infrared laser, the green laser produces more stable molten pool flow with lower velocity, thereby reducing spattering generation.



Download: [Download high-res image \(216KB\)](#)

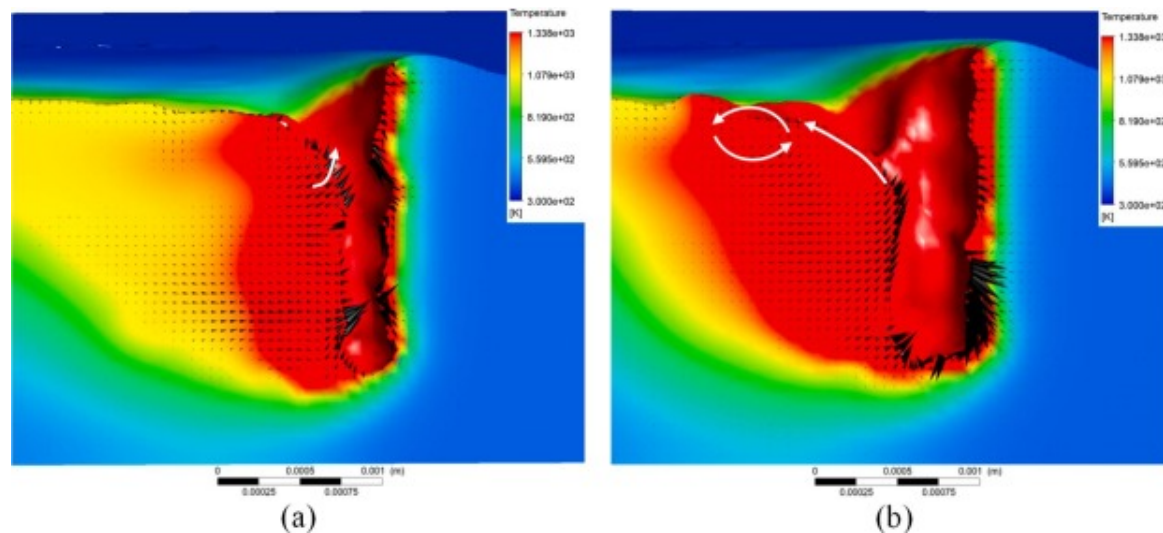
Download: [Download full-size image](#)

Fig. 14. Top surface flow field of molten pool: (a) infrared laser; (b) green laser.

(b) Cross-sectional flow field of molten pool.

Fig. 15 reveals that during ring-core infrared laser welding, the molten pool exhibits a substantial vertical velocity component in its flow field. This elevated velocity enables the

molten material to overcome the spatter formation threshold more readily, thereby promoting spatter generation.



Download: [Download high-res image \(167KB\)](#)

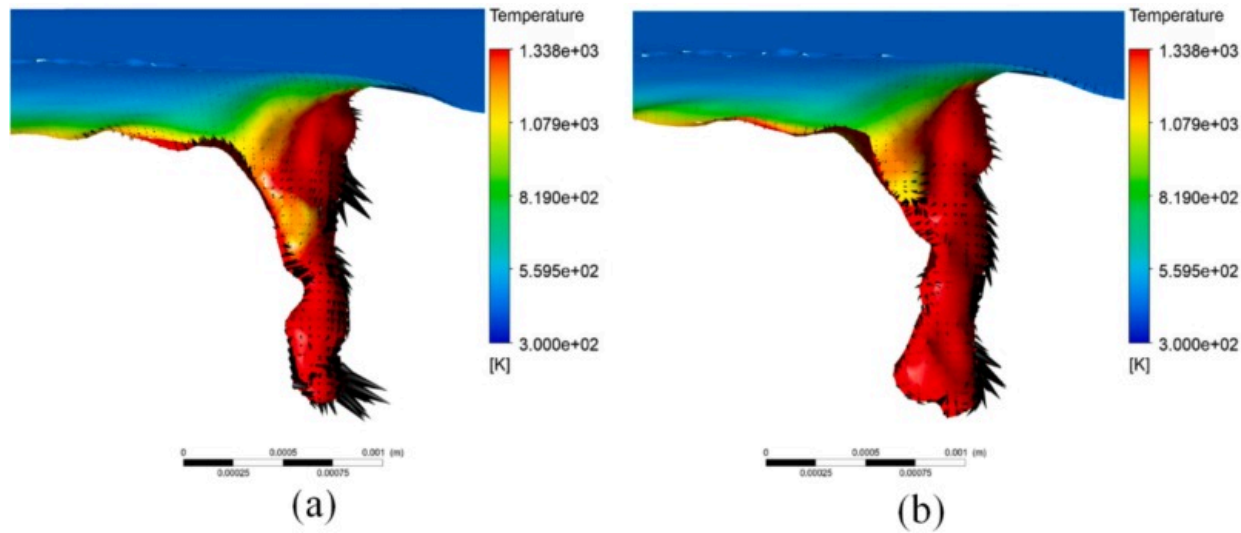
Download: [Download full-size image](#)

Fig. 15. Cross-sectional surface flow field of molten pool: (a) infrared laser; (b) green laser.

Conversely, during ring-core green laser welding, the molten metal flows in an inclined ascent relative to the vertical direction at reduced velocity, while the keyhole wall maintains greater stability. These conditions collectively impede the melt's ability to exceed the spatter threshold. Furthermore, a recirculation flow develops on the molten pool surface, effectively suppressing spatter formation.

4.4. Keyhole region spatter formation and suppression mechanism

As illustrated in Fig. 16, the molten metal flow characteristics in the keyhole zone reveal that compared with infrared laser welding, green laser welding demonstrates less pronounced fluctuations in the keyhole, smaller vertical velocity components in the molten pool, and a wider keyhole opening, which facilitates the escape of metal vapor.

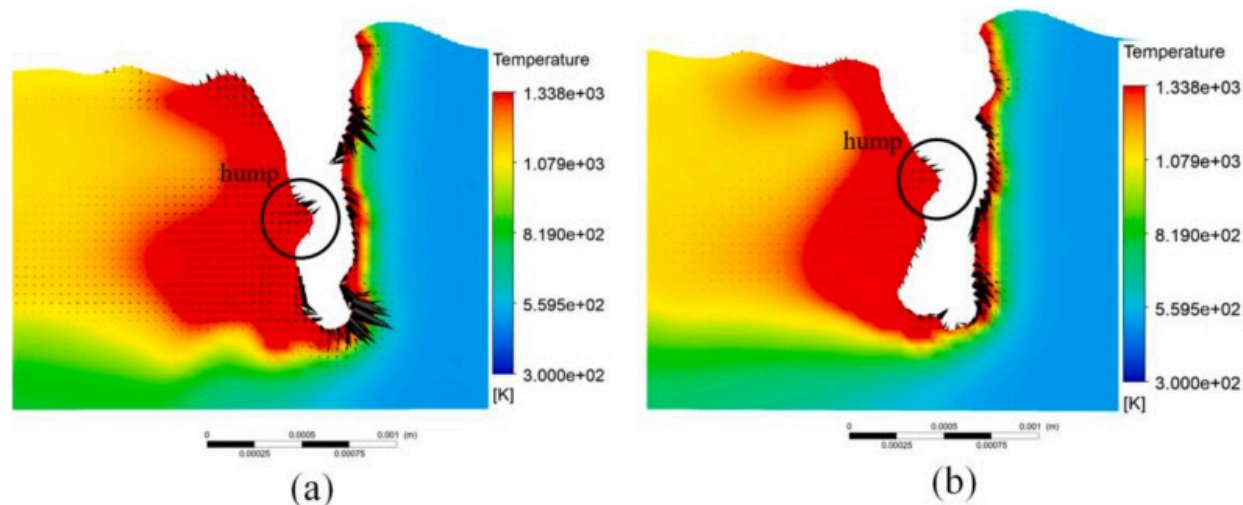


[Download: Download high-res image \(234KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Molten metal flow in the keyhole region: (a) infrared laser; (b) green laser.

As illustrated in Fig. 17, the humps induced by molten pool fluctuations constitute a critical factor in spatter formation. Infrared laser welding exhibits more pronounced keyhole fluctuations, which promote hump generation, while the narrower keyhole results in higher metal vapor pressure. Consequently, when shear forces from metal vapor act upon these humps, they more readily exceed the critical threshold, thus triggering spatter formation.



Download: [Download high-res image \(302KB\)](#)

Download: [Download full-size image](#)

Fig. 17. Humping phenomenon in the keyhole region: (a) infrared laser; (b) green laser.

5. Conclusions

This work systematically investigates the effects of ring-core adjustable (RCA) infrared (1030nm) and green (515nm) lasers on weld surface quality, cross-sectional morphology, mechanical properties, and spatter behavior during T2 pure copper plate welding under varying core-ring energy ratios. Key findings reveal that:

- (1) The wider weld pool produced by the green laser is attributed to a combination of factors, including a larger spot diameter, a lower welding speed, and a higher absorption rate. Compared to infrared laser, the key role of green laser lies in its ability to surpass the keyhole welding threshold in copper more readily.
- (2) Cross-sectional analysis confirms increasing penetration depth and weld width with higher core energy ratios for both processes. Critically, under

non-penetration welding conditions, green laser tends to produce a molten pool with a width-to-depth ratio greater than 1, whereas infrared laser is more likely to generate a pool with a ratio below 1.

- (3) Tensile-shear tests revealed that the green laser exhibited a moderately lower maximum shear force than the infrared laser. For both lasers, the maximum shear force increased with rising core energy ratios, while the shear strength declined after reaching an optimal ratio.
- (4) Per unit time, RCA infrared welding demonstrates both a mass loss rate and a keyhole oscillation frequency roughly 2~3 times higher than those of green laser welding.
- (5) In the molten pool, spatter stems from excessive vertical melt velocity; in the keyhole, it arises from hump detachment under intense vapor shear. Compared to ring-core infrared laser welding, the green laser stabilizes melt flow and enlarges the keyhole aperture. These effects collectively suppress spatter formation by reducing vertical velocity components and attenuating vapor shear.

CRediT authorship contribution statement

Peng Liu: Writing – original draft, Software, Data curation. **Yan Shi:** Writing – review & editing, Resources. **Yang Chen:** Software. **Yuan Liu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The numerical simulation and analysis for this research were conducted at the Laser Processing Laboratory of Changchun University of Science and Technology. Gratitude is extended to Dr. Shi Yan for his valuable guidance. Furthermore, special acknowledgment is given to the Laser Application Center of Trumpf (China) Co. Ltd. and TRUMPF SE + Co. KG (German) for their assistance in the experimental part of this investigation.

Jilin Provincial Scientific and Technological Development Program.

This work was funded by Science and Technology Development Program Project, Jilin Province, China (No: [20230201092GX](#)).

Data availability

Data will be made available on request.

References

- [1] J. de Santiago, H. Bernhoff, B. Ekergr rd, S. Eriksson, S. Ferhatovic, R. Waters, M. Leijon
Electrical motor drivelines in commercial all-electric vehicles: A review
IEEE Trans. Veh. Technol., 61 (2) (2012), pp. 475-484
[View in Scopus](#) [Google Scholar](#)
- [2] S.T. Auwal, S. Ramesh, F. Yusof, S.M. Manladan
A review on laser beam welding of copper alloys
Int. J. Adv. Manuf. Technol., 96 (1–4) (2018)
[Google Scholar](#)
- [3] K.M. Hong, Y.C. Shin
Prospects of laser welding technology in the automotive industry: A review
J. Mater. Process. Technol., 245 (2017), pp. 46-69
 [View PDF](#) [View article](#) [View in Scopus](#) [Google Scholar](#)

- [4] E. Punzel, F. Hugger, R. Dorringer, T.L. Dinkelbach, A. Burger
Comparison of different system technologies for continuous-wave laser beam welding of copper
Int. Conference on Photonic Technol. (2020)
[Google Scholar ↗](#)
- [5] S. Kaierle, L. Overmeyer, C. Hoff, P. Herzog, J. Hermsdorf
Conditioning of copper material surfaces increasing the efficiency of continuous wave laser microwelding
CIRP J. Manuf. Sci. Technol., 14 (2016), pp. 66-70
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] L. Jin, Y. Liang
Fiber laser technologies for photoacoustic microscopy, Visual computing for industry, biomedicine
And Art, 4 (1) (2021), p. 11
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [7] A. Heider, P. Stritt, A. Hess, R. Weber, T. Graf
Process stabilization at welding copper by laser power modulation
Phys. Procedia, 12 (2011), pp. 81-87
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] E.M. Dold, E. Kaiser, K. Klausmann, S. Pricking, S. Zaske, R. Brockmann, High-performance welding of copper with green multi-kW continuous-wave disk lasers, (2019).
[Google Scholar ↗](#)
- [9] S. Engler, R. Ramsayer, R. Poprawe
Process studies on laser welding of copper with brilliant green and infrared lasers
Phys. Procedia, 12 (1) (2011), pp. 339-346

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [10] D.F. Franco, Wobbling laser beam welding of copper, Universidade NOVA de Lisboa (Portugal), 2017.

[Google Scholar ↗](#)

- [11] R.B. A, D.H.A. B, A.F.L.a. b, A.M. B, Tailored absorptivity: Increasing the laser weldability of copper through surface structuring - ScienceDirect, Materials Letters 283 (2020).

[Google Scholar ↗](#)

- [12] G. Nordet, C. Gorny, Y. Mayi, J. Daligault, M. Dal, A. Effernelli, E. Blanchet, F. Coste, P. Peyre
Absorptivity measurements during laser powder bed fusion of pure copper with a 1 kW cw green laser
Optics Laser Technol., 147 (2022), Article 107612-

[Google Scholar ↗](#)

- [13] A. Ganser, T. Eder, M.F. Zäh
Laser welding of copper using a high power disc laser at green wavelength
Procedia CIRP (2018)

[Google Scholar ↗](#)

- [14] S.G. A, L.T. A, M.F.Z. A, Laser beam welding of copper foil stacks using a green high power disk laser, Procedia CIRP 94 (2020) 582-586.

[Google Scholar ↗](#)

- [15] A. Hess, R. Schuster, A. Heider, R. Weber, T. Graf
Continuous wave laser welding of copper with combined beams at wavelengths of 1030 nm and of 515 nm
Phys. Procedia, 12 (2011), pp. 88-94

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [16] S.C. Li, G.Y. Chen, S. Katayama, Y. Zhang
Relationship between spatter formation and dynamic molten pool during high-power deep-penetration laser welding
Appl. Surf. Sci., 303 (2014), pp. 481-488
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [17] W.C. Li, H. Dong, B. Zhang, S.Y. Zou, W.D. Mu, Y. Cai
The influence of adjustable ring-mode laser on the formation of intermetallic compounds and mechanical properties of ultra-thin Al-Cu lap welded joints
J. Mater. Process. Technol., 332 (2024), p. 14
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [18] L. Wang, X.D. Gao, F.R. Kong
Keyhole dynamic status and spatter behavior during welding of stainless steel with adjustable-ring mode laser beam
J. Manuf. Process., 74 (2022), pp. 201-219
 [View PDF](#) [View article](#) [Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [19] L. Wang, M. Mohammadpour, B.X. Yang, X.D. Gao, J.P. Lavoie, K. Kleine, F.R. Kong, R. Kovacevic
Monitoring of keyhole entrance and molten pool with quality analysis during adjustable ring mode laser welding
Appl. Optics, 59 (6) (2020), pp. 1576-1584
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [20] L. Wang, M. Mohammadpour, X.D. Gao, J.P. Lavoie, K. Kleine, F.R. Kong, R. Kovacevic
Adjustable ring mode (ARM) laser welding of stainless steels
Opt. Lasers Eng., 137 (2021), p. 12
 [View PDF](#) [View article](#) [Google Scholar ↗](#)
- [21] Y. Qi, G.Y. Chen, D.Z. Liu

Droplet spatter suppression in laser lap welding of galvanized sheets using additional coaxial annular laser source

Opt. Laser Technol., 149 (2022), p. 7

[Google Scholar ↗](#)

- [22] M.R. Maina, Y. Okamoto, A. Okada, M. Närhi, J. Kangastupa, J. Vihinen
High surface quality welding of aluminum using adjustable ring-mode fiber laser

J. Mater. Process. Technol., 258 (2018), pp. 180-188



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [23] Q.H. Li, M.L.L. Luo, Z.Y. Mu, A.G. Huang, S.Y. Pang
Improving laser welding via decreasing central beam density with a hollow beam

J. Manuf. Process., 73 (2022), pp. 939-947



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [24] J.v.a. b, f.v.a. c,
Impact of multi-focus beam shaping on the process stability

Opt. Laser Technol., 112 (2019), pp. 278-283

[Google Scholar ↗](#)

- [25] S. Liebl, R. Wiedenmann, A. Ganser, P. Schmitz, M.F. Zaeh
Laser welding of copper using multi mode fiber lasers at near infrared wavelength

Phys. Procedia, 56 (2014), pp. 591-600



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [26] A.F.H.K. Powell
Spatter in laser welding
J. Laser Appl., 23 (3) (2011), pp. 3337-3344

[Google Scholar ↗](#)

- [27] C. Börner, T. Krüssel, K. Dilger
Process characteristics of laser beam welding at reduced ambient pressure
Proc. SPIE Int. Soc. Opt. Eng., 8603 (2013)
[Google Scholar ↗](#)
- [28] A.F.H. Kaplan, J. Powell
Spatter in laser welding
J. Laser Appl., 23 (3) (2011), p. 7
[Google Scholar ↗](#)
- [29] M. Hummel, M. Külkens, C. Schler, W. Schulz, A. Gillner
In situ X-ray tomography investigations on laser welding of copper with 515 and 1030 nm laser beam sources
J. Manuf. Process. (2021)
[Google Scholar ↗](#)
- [30] S. Huang, R.D. Lu, M. Lou, T.L. Lv, J.H. Yao, Y.B. Li
Effect of oscillation parameters on adjustable-ring mode (ARM) laser beam welding of aluminum alloys
J. Manuf. Process., 113 (2024), pp. 307-318
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[View Abstract](#)

Recommended articles

Based on reading popularity

Research article Full text access

Conditioning of copper material surfaces increasing the efficiency of continuous wave laser microwelding

Stefan Kaierle, ..., Jörg Hermsdorf

CIRP Journal of Manufacturing Science and Technology • Volume 14 • 2016 • Pages 66-70



[View PDF](#)

Research article Open access

Study of spatter net forming mechanism and penetration mode under flexible ring mode laser welding

Yang Zhang, ..., Kejin Zhang

Journal of Materials Research and Technology • Volume 24 • 2023 • Pages 2213-2225



[View PDF](#)

Research article Open access

Influence of dynamic beam shaping on the geometry of the keyhole during laser beam welding

Jonas Wagner, ..., Thomas Graf

Procedia CIRP • Volume 111 • 2022 • Pages 448-452



[View PDF](#)

Research article Full text access

Achieving high robust laser conduction welding and enhanced joint conductivity in pure copper foil stacks using a flat-top blue laser

Le Wan, ..., Hongze Wang

Optics & Laser Technology • Volume 174 • 2024 • Article 110701



[View PDF](#)

Research article Open access

A hybrid sharp–diffuse interface approach to accurately model melt pool dynamics with rapid evaporation in laser-based processing of metals

Nils Much, ..., Magdalena Schreter-Fleischhacker

Computer Methods in Applied Mechanics and Engineering • Volume 457 • 2026 • Article 119023



[View PDF](#)

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

Mendeley Readers

2

[View details ↗](#)

© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.